

Trainings ▾

Preparatory Steps

With EGR

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the air intake piping and intake manifold. Refer to Procedure 010-023 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-023-tr.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)
- Remove the engine brakes, if applicable. Refer to Procedure 020-024 in Section 20.

(/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)

- Remove the engine brake oil feed line, if equipped.
- Remove the locking capscrew from the Metri-Pack™ 12-pin connector at the front of the rocker lever housing, and separate the connector.
- Remove the internal engine wiring harness holding clamp capscrews.
- Disconnect the Deutsch™ 2-pin connectors from the internal engine wiring harness to the fuel injectors.
- Disconnect the internal engine wiring harness from the engine.
- Remove the push rods or push tubes. Refer to Procedure 004-014 in Section 4.

(/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)

- Remove the rocker lever assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html)
- Remove the crossheads. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html)
- Remove the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/35/35-006-026-tr.html)
- Remove the air compressor air supply tube from the exhaust gas recirculation (EGR) mixer. Refer to Procedure 012-109 in Section 12.
(/qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html)

- Remove the EGR differential pressure sensor harness and sensor. Refer to Procedure 019-370 in Section 19.
(/qs3/pubsys2/xml/en/procedures/88/88-019-370.html)
This procedure is found in the CM870 Electronic Control System, ISM Engines Troubleshooting and Repair Manual, Bulletin 4021381.

- Disconnect the exhaust pressure sensor from the harness. Refer to Procedure 019-376 in Section 19.
(/qs3/pubsys2/xml/en/procedures/88/88-019-376.html)
This procedure is found in the CM870 Electronic Control System, ISM Engines Troubleshooting and Repair Manual, Bulletin 4021381.
- Disconnect the EGR differential pressure sensor. Refer to Procedure 011-026 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)

- Disconnect the EGR exhaust pressure sensor tubes. Refer to Procedure 011-027 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)

- Remove the EGR mixer and air horn assembly. Refer to Procedure 011-021 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html)

- Remove the EGR connection tube. Refer to Procedure 011-025 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)

- Remove the water manifold. Refer to Procedure 008-061 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)

- Remove the EGR cooler vent lines. Refer to Procedure 011-031 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)

- Remove the EGR cooler coolant return and supply tubes. Refer to Procedure 011-019 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-019-tr.html)

- Remove the EGR valve. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)

- Remove the EGR cooler. Refer to Procedure 011-019 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-019-tr.html)

- Disconnect the cylinder head vent lines.

- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3.

(/qs3/pubsys2/xml/en/procedures/35/35-003-013-tr.html)

- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-007-tr.html)

- Remove the fuel lines from the cylinder head. Refer to Procedure 006-024 in Section 6.

(/qs3/pubsys2/xml/en/procedures/35/35-006-024-tr.html)

- Remove the fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-036-tr.html)
- Remove the fan. Refer to Procedure 008-040 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-040-tr.html)
- Disconnect the engine control module (ECM) connectors from the ECM. Refer to Procedure 006-006 in Section 6. (/qs3/pubsys2/xml/en/procedures/35/35-006-006-tr.html)
- Remove the CM870 ECM mounting bracket. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/88/88-019-031.html) This procedure is found in the CM870 Electronic Control System, ISM Engines Troubleshooting and Repair Manual, Bulletin 4021381.
- Remove the refrigerant compressor and mounting bracket.

Without EGR

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- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Remove the air intake piping and intake manifold. Refer to Procedure 010-023 in Section 10.

(/qs3/pubsys2/xml/en/procedures/35/35-010-023-tr.html)

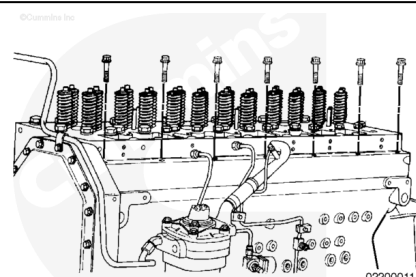
- Remove the crankcase breather, if applicable. Refer to Procedure 003-001 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-001-tr.html)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Remove the engine brakes, if applicable. Refer to Procedure 020-024 in Section 20.
(/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)
- Remove the locking capscrew from the Metri-Pack™ 12-pin connector at the rear of the rocker lever housing, and separate the connector.
- Remove the internal engine wiring harness holding clamp capscrews.
- Remove the locking capscrew from the Deutsch™ 2-pin connectors from the internal engine wiring harness to the fuel injectors.
- Remove the internal engine wiring harness from the engine.
- Remove the push rods or push tubes. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)
- Remove the rocker lever assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html)
- Remove the crossheads. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html)
- Remove the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/35/35-006-026-tr.html)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-013-tr.html)

- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-007-tr.html)
- Remove the fuel lines and air compressor tube from the cylinder head.
- Remove the fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-036-tr.html)
- Remove the cooling fan. Refer to Procedure 008-040 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-040-tr.html)
- Remove all the necessary brackets and clamps.

Remove

⚠ WARNING ⚠

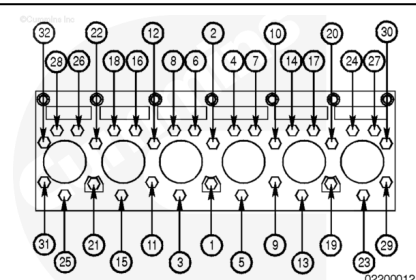
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the seven 12-point cylinder head capscrews on the fuel pump side of the engine.

Remove the remaining 32 cylinder head capscrews in the sequence shown.

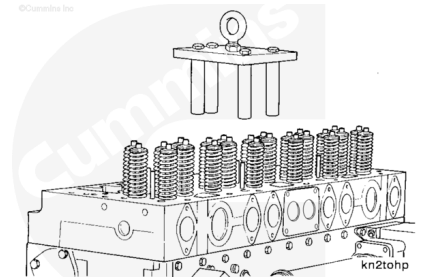
Three capscrews are located inside the intake ports.



Install the cylinder head lifting bracket, Part Number 3822479, with two rocker lever support mounting capscrews and two of the long rocker housing mounting capscrews.

Tighten the capscrews.

Torque Value: 47 n•m [35 ft-lb]

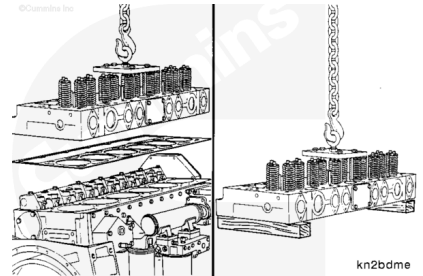


⚠ CAUTION ⚠

Place the cylinder head on wood blocks to prevent damage to the combustion face.

Use a hoist or hydraulic arm to remove the cylinder head.

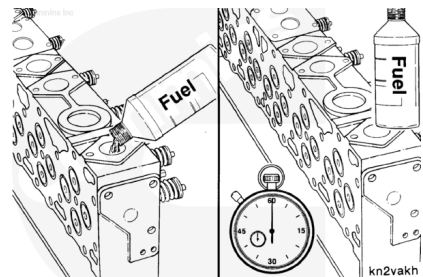
Remove and discard the cylinder head gasket.



Leak Test

Inspect the valves for indications of leaking or burning. If indications of leaking or burning are found, the valves and the seats **must** be resurfaced.

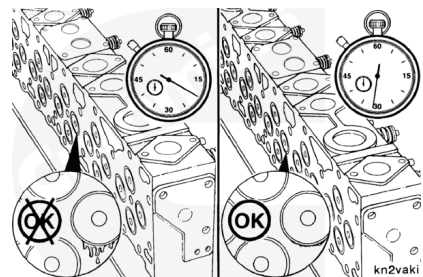
Test the cylinder head for damage. Set the cylinder head down with the exhaust ports facing up. Pour fuel into one of the exhaust ports until it is full. Set the container of fuel down, and start a timer.



If a drop of fuel drips or runs down the face of the cylinder head within 30 seconds, the exhaust valves and the seats **must** be resurfaced.

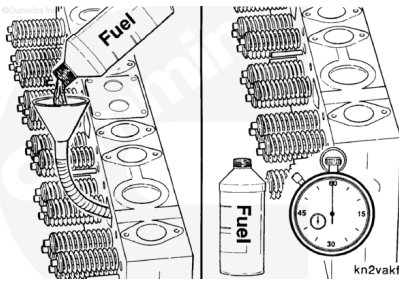
If a drop of fuel has **not** run down the face of the cylinder head in 30 seconds, the exhaust valves are acceptable.

Repeat the process for all six cylinders.



Pour fuel into one of the intake ports until it is full. Set the container of fuel down, and start the timer.

Each intake port supplies four intake valves (two cylinders) on M Series cylinder heads.

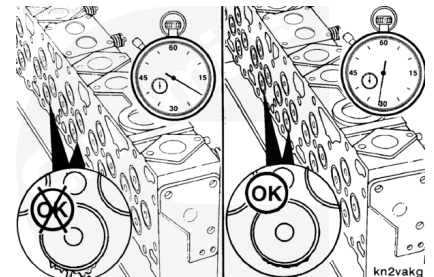


If a drop of fuel drips or runs down the face of the cylinder head within 30 seconds, the intake valves and the seats **must** be resurfaced.

If a drop of fuel has **not** run down the face of the cylinder head in 30 seconds, the intake valves are acceptable.

Repeat the process for all six cylinders.

If any of the intake or exhaust valves fail the test, all of the valves and valve seats **must** be resurfaced.

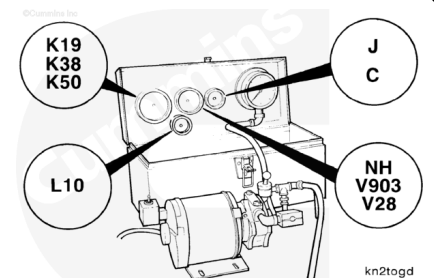


Vacuum Test

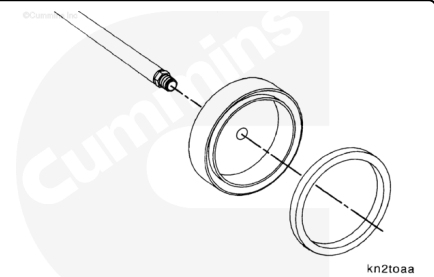
Use the vacuum tester, Part Number ST-1257, to inspect the seal between the valve and the valve seat.

The valve vacuum tester is **not** recommended for use on used cylinder heads. See the Leak Test section of this procedure for the recommended procedure to check used cylinder heads.

The valve vacuum tester can be used to test all Cummins® engine models. The seal ring, Part Number ST-1257-35, and vacuum cup, Part Number 3376100, are used on the M11 cylinder heads.



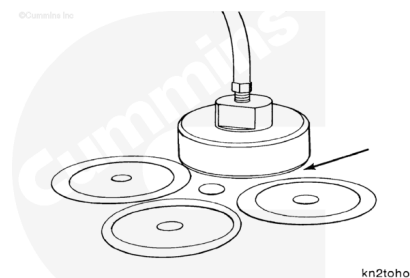
Install the seal ring and vacuum cup to the vacuum line hose.



The valves and valve seats **must** be clean and dry when vacuum testing.

Cover the valve with the cup and seal.

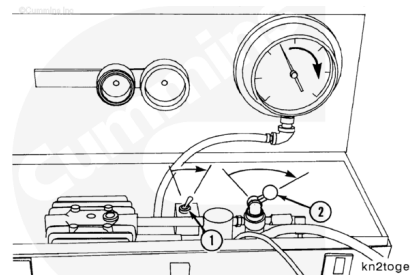
The seal **must** make a tight contact on the cylinder head around the valve.



Move the toggle switch (1) to the ON position.

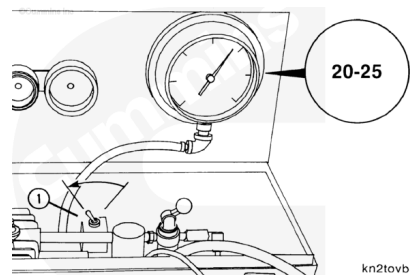
Turn the vacuum control valve (2) to the OPEN position.

The vacuum control valve is in the OPEN position if the vacuum gauge needle moves **clockwise**.



Operate the vacuum pump until the gauge indicates the specified vacuum.

Valve to Valve Seat Vacuum		
mm-hg		in-hg
508	MIN	20
635	MAX	25

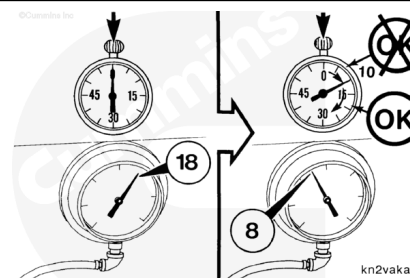


Turn the toggle switch to the OFF position.

Use a stopwatch and start the timing when the needle on the gauge indicates 457 mm Hg [18 in Hg].

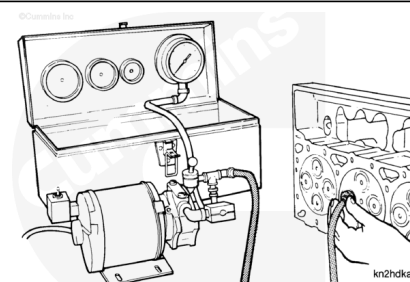
Stop the timing when the needle on the gauge indicates 203 mm Hg [8 in Hg].

The elapsed time for the needle to move between the specified gauge readings **must** be 10 seconds or more.



If the elapsed time is less than 10 seconds, perform the following checks:

- Repeat the test to be sure the equipment is operating properly.
- Use a mallet to hit the valve stem lightly to make sure the valve is seated. Repeat the test.
- Apply a thin layer of grease on the outside diameters of the insert and the valve head. Repeat the test. The grease pattern will show the point of leakage.

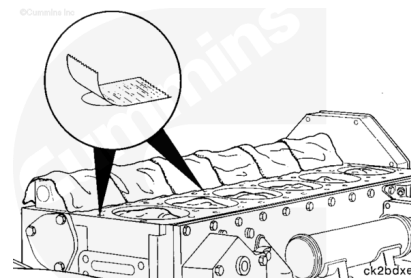


A break in the grease seal pattern will indicate leakage between the valves and valve seat or the valve seat insert and the cylinder head.

Clean

⚠ CAUTION ⚠

If the oil passages are not plugged or covered before cleaning, a crankshaft failure can occur.

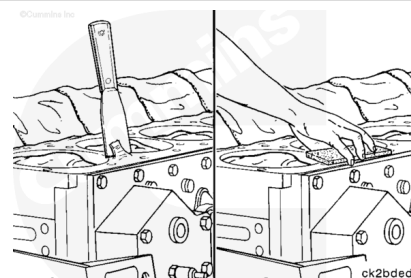


Place clean, lint-free rags over the cam followers and above the pistons to prevent dirt from falling into the engine.

Plug or cover the coolant and oil passages in the cylinder block deck.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a gasket scraper to clean the cylinder block deck surface.

Use Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent, and solvent to remove any residual gasket material from the cylinder block deck surface.

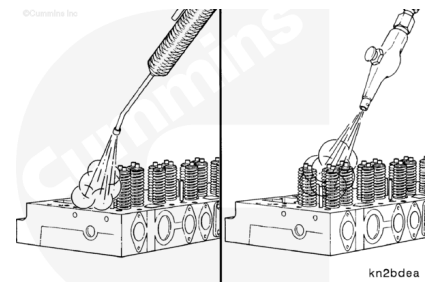
Make sure the cylinder head capscrew holes are clean and free of debris, oil, and coolant.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

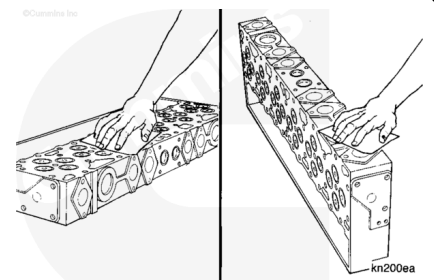
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Steam clean the cylinder head and dry with compressed air.
Use compressed air to blow out all capscrew holes.

⚠ WARNING ⚠

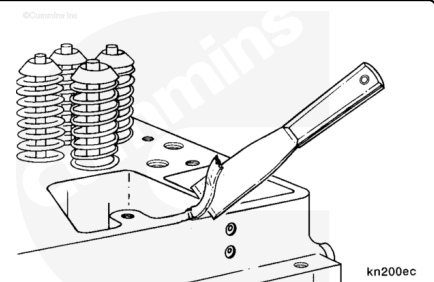
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendation for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent, and solvent to clean the cylinder head combustion face.

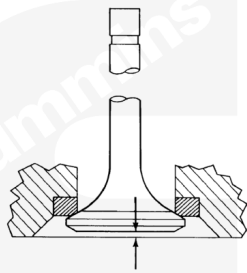
Clean the exhaust manifold gasket surface.

Clean the rocker housing gasket surface.



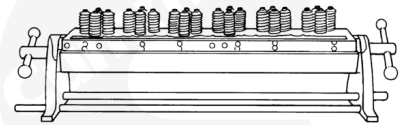
Disassemble

Prior to disassembly, measure and record the valve recess and valve tip height above the top deck of the cylinder head. Refer to the Clean and Inspect for Reuse section in this procedure.



kn2istb

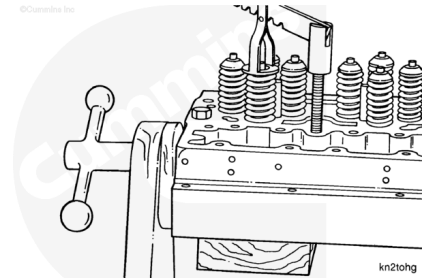
Install the cylinder head in the head holding fixture, Part Number ST-583, as shown.



kn2tohf

Install a wood block between the valves and head holding fixture to support the valves.

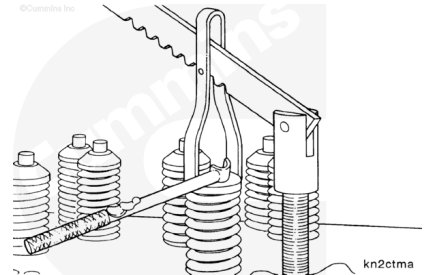
Use the valve spring compressor, Part Number ST-448, the compressor threaded adapter, Part Number 3376850, and the compressor adapter, Part Number 3376851, to compress the valve springs.



kn2tohg

Use a pencil magnet to remove the valve collets. Discard the valve collets.

Slowly release the pressure on the valve spring.

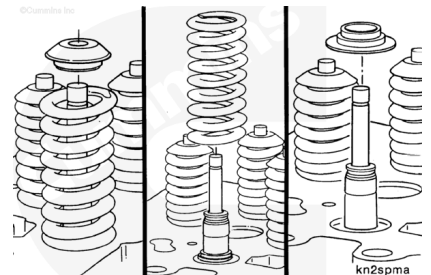


kn2ctma

Remove the valve spring retainer.

Remove the valve spring.

Remove the valve spring wear plate.



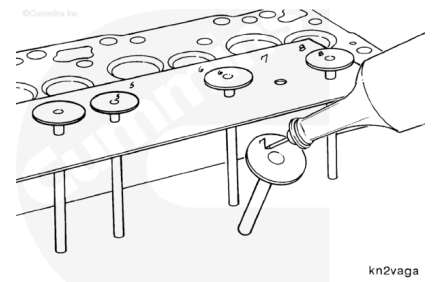
kn2spma

Remove the valve.

Mark the valves with an electric pencil (engraving tool) for location as they are removed.

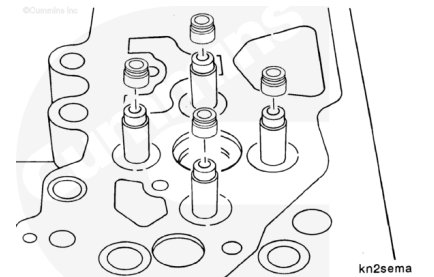
The intake and exhaust valves are manufactured from different materials and are **not** interchangeable.

Repeat the procedure to remove the remaining valve collets, retainers, springs, wear plates, and valves.

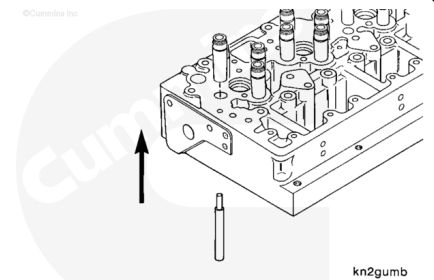


Remove the valve seals from the valve guides.

Prior to removing the valve guides, measure and record the valve guide inside diameter and height. Refer to the Clean and Inspect for Reuse section of this procedure.

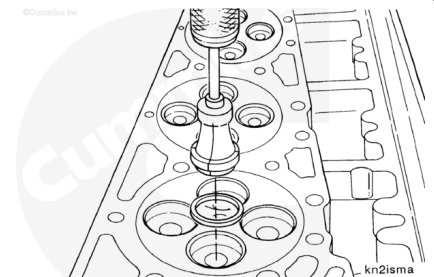


Use valve guide driver, Part Number 3376398, to remove the old valve guides.



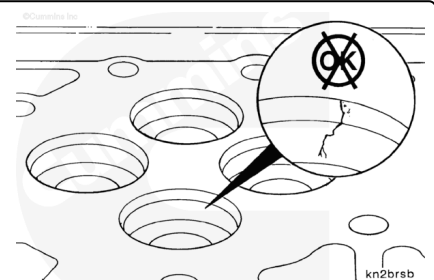
Prior to removing the valve seat inserts, refer to the Leak Test and the Inspect for Reuse sections of this procedure. The condition of the valve, the amount of recess, and the sealing of the valve on the seat insert all help determine whether or **not** a seat insert needs to be replaced.

Use slide hammer assembly, Part Number 3376616, and valve seat extractor, Part Number 3376146, to remove the valve seat inserts from the cylinder head.



Inspect the insert bore for cracks or other damage.

If cracks or other damage is found, it is possible to repair the cylinder head by machining the insert bore for oversize valve seat inserts. Reference the Alternative Repair Manual, Bulletin 3710310.

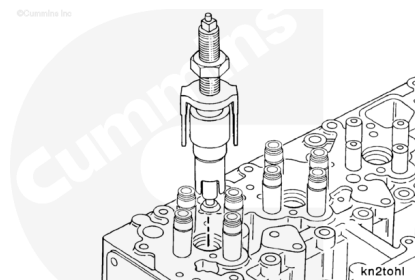


Perform the coolant passage and fuel passage pressure test before removing the injector sleeves.

Use injector sleeve puller, Part Number 3377079, to remove the injector sleeves from the cylinder head.

Follow the directions supplied with the tool to remove the sleeves.

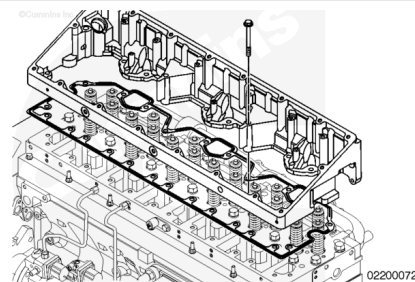
Remove the injector sleeve o-ring from the injector bore in the cylinder head. Discard the o-ring.



Repair

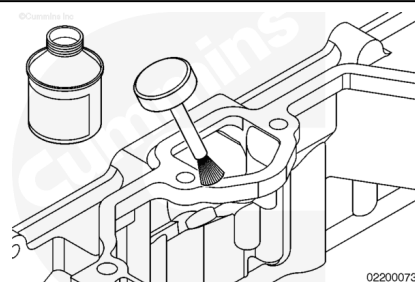
Head erosion around intake port - If **not** already removed, remove the rocker lever housing. Refer to Procedure 003-013 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-013-tr.html)

If **not** already removed, remove the rocker lever housing gasket.

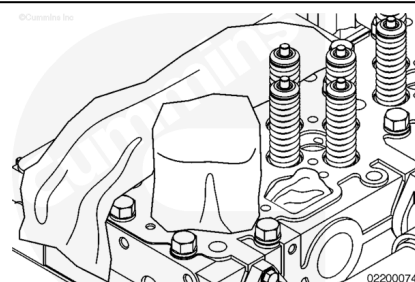


Use Repair Kit, Part Number 4918684, when repairing the cylinder head erosion around the intake port.

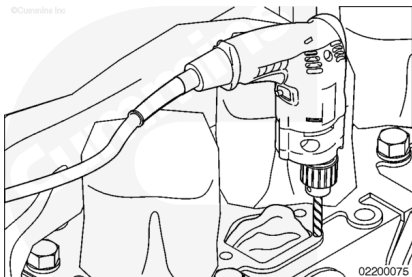
Clean the mating face of the rocker lever housing with LPS™ ZeroTri™, or equivalent. Denatured alcohol is also acceptable. Apply Belzona™ Release Agent 9411 around the corresponding intake port with Belzona™ 1311 Ceramic R-Metal. The release agent needs to be applied 15 to 20 minutes prior to contact with the Belzona™ 1311 Ceramic R-Metal.



Plug the intake ports and cover the valve springs and cam followers to prevent debris from entering the engine.



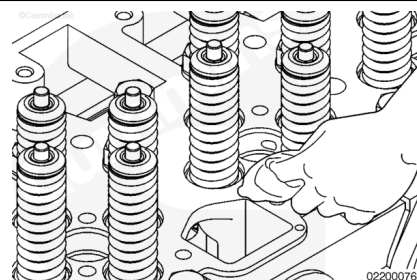
Drill out the eroded sections of the cylinder head, 0.381 mm to 1.02 mm [0.015 in to 0.040 in] deep for the filler material with an Irwin™ jetpoint drill bit.



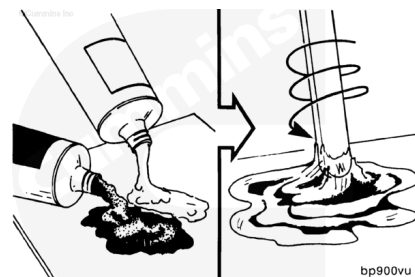
Remove the drill chips from the cylinder head area.

Remove the intake plugs from the ports.

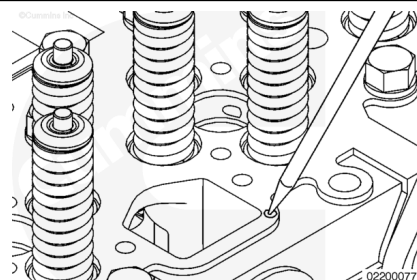
Clean any oil or oil residue away from the gasket surface and the drilled pockets in the cylinder head. Use LPS™ ZeroTri™, or equivalent. Denatured alcohol is also acceptable. Make sure there is no oily residue.



Mix up three parts Belzona™ 1311 Base to one part Belzona™ Solidifier. It needs to be mixed thoroughly. The color needs to be uniform with no light colored streaks present.



When the surface is clean and dry, apply the mixed Belzona™ 1311 to the drill pockets in the cylinder head. Make sure there is plenty of material to fill the drilled area. Press the filler into the cavity repeatedly to make sure that no air pockets remain.



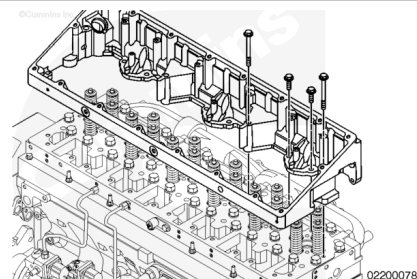
Install the rocker lever housing **only** to the cylinder head. Do **not** use a gasket.

Install the rocker housing bolts in the four corners around the port being repaired. The rocker housing is being used to form the liquid metal to the cylinder head surface.

Tighten the rocker housing bolts.

Torque Value: 61 n•m [45 ft-lb]

Allow the Belzona® to set up for at least 3 hours.



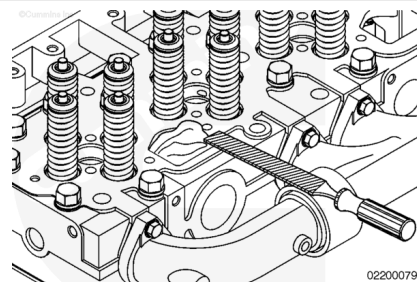
The set up time is temperature dependant.

Temperature		Required Setup Time (Hours)
°C	[°F]	
Below 16	Below [60]	3
16 to 21	[60 to 70]	2 1/2
21 to 27	[70 to 80]	2
27 to 32	[80 to 90]	1 1/2
32 and above	[90 and above]	1

Replace the intake port plugs to prevent debris from entering the cylinder head intake port.

Clean and remove the excessive Belzona® with your finger or a file. Do **not** allow Belzona® or any debris to fall into the intake port. Blend or smooth the material around the repaired area with a file or abrasive block.

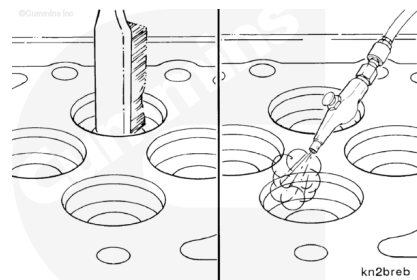
Remove the intake plugs and any other covers.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

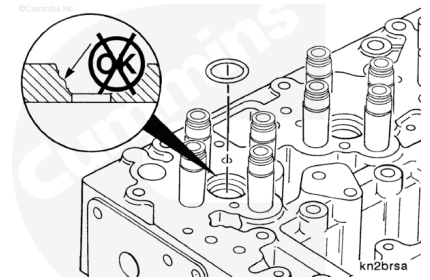
Use a wire brush and solvent to clean the deposits from the valve seat insert bores.

Use solvent to clean the cylinder head.

Dry with compressed air.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the injector bore.

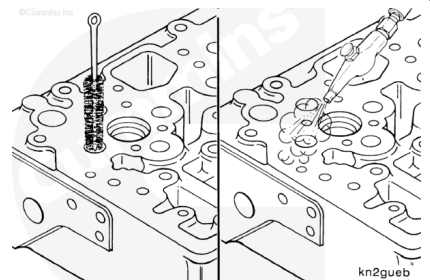
Dry with compressed air.

Inspect the injector seat surface in the bottom of the injector bore.

If the bead in the cylinder head is **not** smooth, the injector bore **must** be repaired. Reference the L10 Alternative Repair Manual, Bulletin 3810310.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



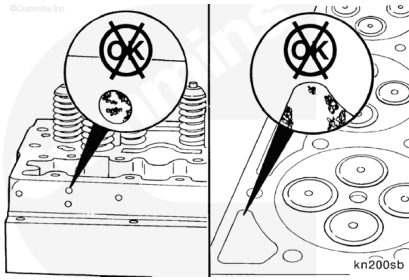
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a flexible brush and solvent to clean the valve guide bores in the cylinder head. Dry with compressed air.

Inspect the fuel drillings and water passages for restrictions or foreign material.

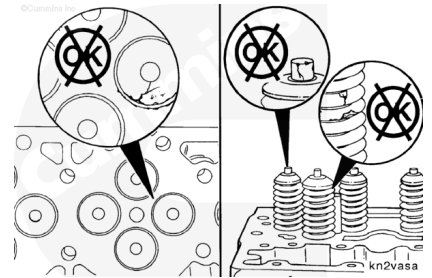
Remove any obstructions.



Inspect the valves and valve springs for cracks, bent or broken valve stems, broken valve spring collets, or other damage.

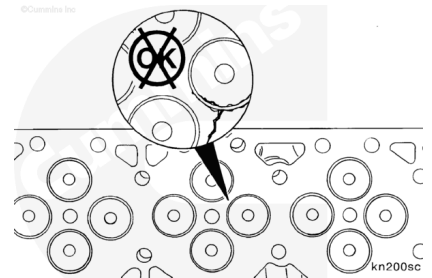
Inspect the valves for indications of leaking or burning.

If cracked or damaged parts, or indication of leaking or burning is found, the cylinder head **must** be rebuilt. Reference the M11 Series Shop Manual, Bulletin 3666075.



Inspect the cylinder head and valves for cracks or other damage.

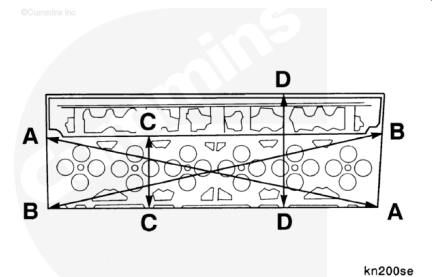
If a crack in the cylinder head is suspected, pressure-test the cylinder head. See the Pressure Test section of this procedure.



Measure the flatness of the cylinder head combustion face surface as follows:

- AA and BB (corner to corner)
- CC (across combustion face)
- DD (across entire cylinder head surface).

Dimensions CC and DD **must** be checked from front to rear of the cylinder head.

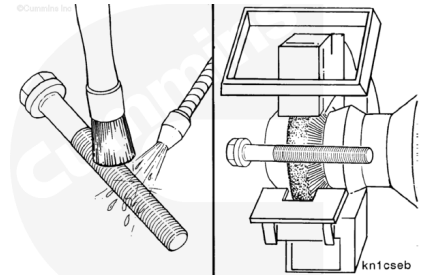


Cylinder Head Flatness				
	mm		in	
AA and BB		0.200	MAX	0.008
CC		0.076	MAX	0.003
DD		0.127	MAX	0.005

If the cylinder head is pitted, has grooves or wear greater than the maximum specified, the cylinder head surface **must** be machined or cut. Reference the L10 Alternative Repair Manual, Bulletin 3810310.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendation for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

Do not use caustic or acid solutions to clean the cylinder head capscrews to avoid damage to the cylinder head capscrews.

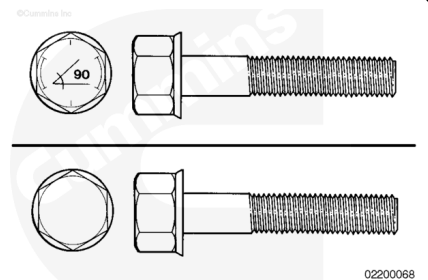
Clean the cylinder head capscrews with a petroleum-based solvent.

Clean the capscrews thoroughly with a wire brush, a wire wheel (soft), or a nonabrasive bead blast to remove deposits from the shank and threads.

The use of flange cylinder head capscrews with the torque plus angle method of installation places the capscrews beyond the yield point and permanently stretches the capscrews. These capscrews can be reused throughout the life of the engine, unless the capscREW exceeds the specified free length. The free length **must** be checked to avoid bottoming in the block during installation.

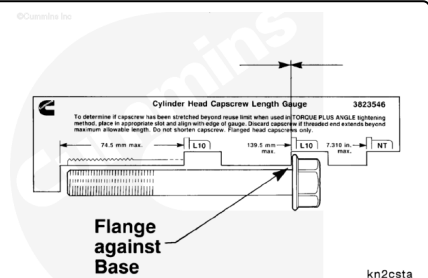
Cylinder head capscREW length gauge, Part Number 3823546, has been designed to check capscREW free length.

Note : The new capscrews are coated with grey coating and do not have the 90 degree markings on the capscREW.

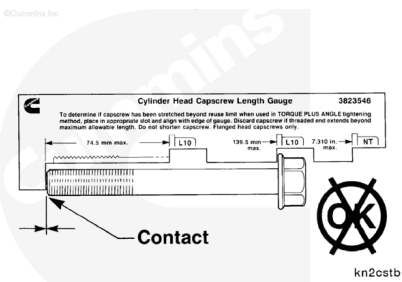


To check the capscREW free length, place the cylinder head of the capscREW in the appropriate slot, long or short, with the flange against the base of the slot.

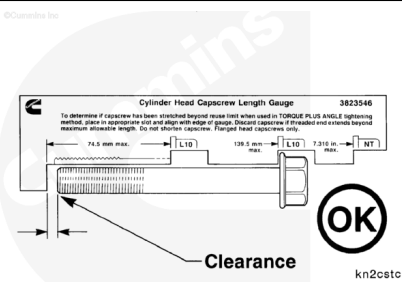
Use the slot marked for L10 cylinder head capscREWS when checking M11 cylinder head capscREWS.



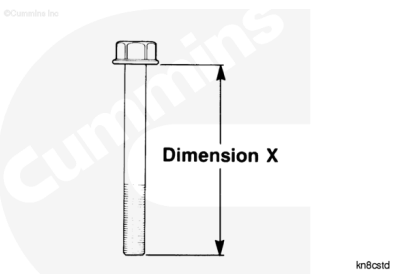
If the end of the capscrew touches the foot of the gauge, the capscrew is too long and **must** be discarded.



If there is clearance between the end of the capscrew and the bottom base of the tool, the capscrew can be reused.



The capscrew can also be checked using a set of calipers. The maximum allowable free length is measured from the bottom of the flange to the end of the capscrew (dimension X).

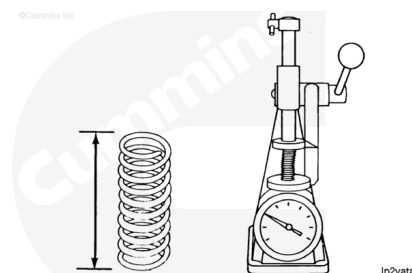


Cylinder Head Capscrew Free Length				
	mm		in	
304584 9 Short Intake Port Capscrews		74.5	MAX	2.933
304585 0 Long Hex Head Capscrews		139.5	MAX	5.492
492318 7 Long Hex Head Capscrews		139.5	MAX	5.492

Use valve spring tester, Part Number 3375182, to measure the valve spring load at the valve spring working height.

Valve Spring Load at 71.5 mm [2.814 in]

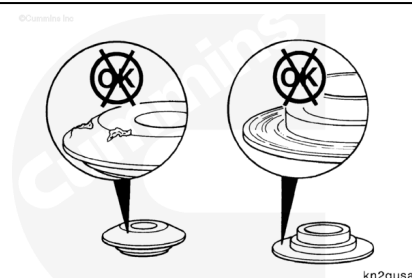
n		lbf
1252	MIN	281
1584	MAX	356



If the valve spring load for the working height is less than the minimum specified, the valve spring **must** be replaced.

Inspect the valve spring retainers and valve spring guides for damage or work areas.

Discard damaged and worn parts.

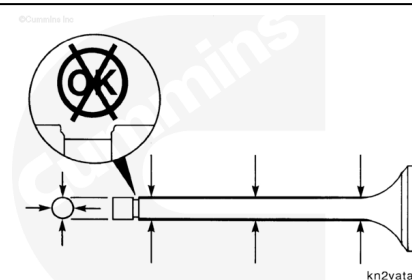


Inspect the valves for damage and the collet grooves for wear.

Measure the outside diameter of the valve stem.

Valve Stem Outside Diameter (OD)

mm		in
9.580	MIN	0.3772
9.633	MAX	0.3793



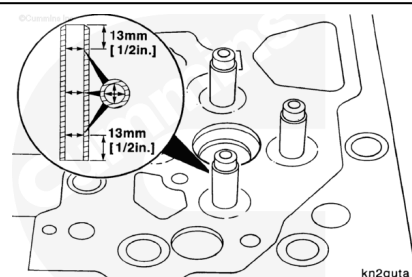
If the valves are damaged or the stems are worn smaller than the minimum specified, the valves **must** be replaced.

Inspect the valve guides, if **not** removed in the Disassemble section, for chips or cracks.

Measure the inside diameter of the valve guides in locations 13 mm [0.5 in] from each end and at the center as shown.

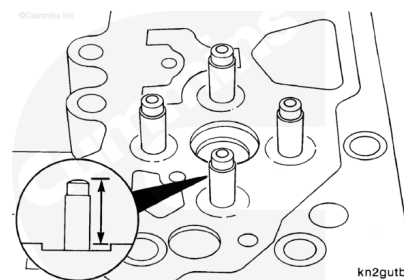
Used Valve Guide Inside Diameter (Installed)

mm		in
9.663	MIN	0.380
9.713	MAX	0.382



If the valve guide inside diameter is **not** within specification, the valve guide **must** be replaced.

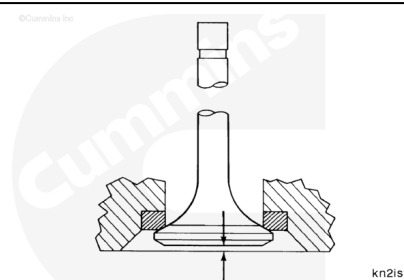
Measure the valve guide installed height above the cylinder head top deck surface.



Valve Guide Height (Installed)			
	mm		in
Short Guides	27.15	MIN	1.069
	27.65	MAX	1.089
Tall Guides	46.32	MIN	1.824
	46.58	MAX	1.834

Install a new or reconditioned valve in the valve guide. Hold the valve firmly against the valve seat insert.

Use a depth gauge, Part Number 3823495, to measure the valve recess in the cylinder head.



Valve Recess for Cylinder Heads Built Prior to August 21, 2006

mm		in
0.76	MIN	0.030
1.17	MAX	0.046

Maximum Operational Valve Recess for Cylinder Heads Built Prior to August 21, 2006

mm		in
3.05	MAX	0.120

Valve Recess for Cylinder Heads Built After August 21, 2006, with the Scallop

mm		in
0.51	MIN	0.020
0.92	MAX	0.036

Maximum Operational Valve Recess for Cylinder Heads Built After August 21, 2006, with the Scallop

mm		in
2.80	MAX	0.110

Maximum Valve Recession Variation Between Valves
on Same Cylinder for Cylinder Heads Built Prior to
August 21, 2006

mm		in
0.76	MAX	0.030

Maximum Valve Tip Height for Cylinder Heads Built
Prior to August 21, 2006

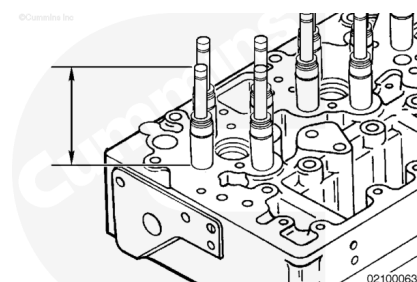
mm		in
99.60	MAX	3.921

Maximum Valve Recession Variation Between Valves
on Same Cylinder for Cylinder Heads Built After
August 21, 2006, with the Scallop

mm		in
0.51	MAX	0.020

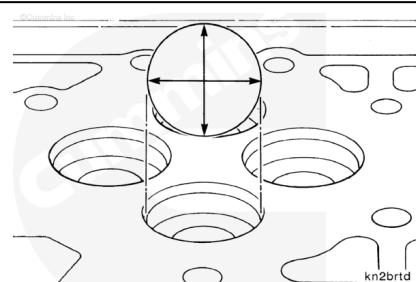
Maximum Valve Tip Height for Cylinder Heads Built
After August 21, 2006, with the Scallop

mm		in
99.35	MAX	3.911



Measure the inside diameter of the valve seat insert bore in the cylinder head.

Insert Bore Inside Diameter (ID)		
mm		in
45.920	MIN	1.8079
45.935	MAX	1.8085

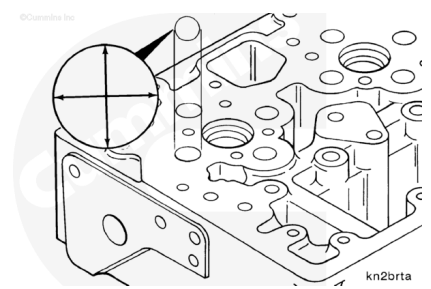


If the valve seat insert bore inside diameter does **not** meet the specifications given, reference the L10 Alternative Repair Manual, Bulletin 3810310 for the oversize valve installation insert installation.

Measure the inside diameter of the valve guide bore in the cylinder head.

Valve Guide Bore Inside Diameter (ID)		
mm		in
16.480	MIN	0.6488

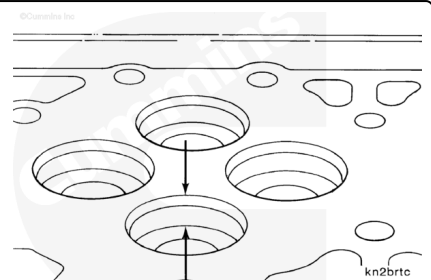
Valve Guide Bore Inside Diameter (ID)		
mm		in
16.500	MAX	0.6496



If the valve guide bore is worn larger than the maximum specified, the valve guide bore can be machined, and a 0.51 mm [0.02 in] oversize valve guide installed. Refer to the L10 Alternative Repair Manual, Bulletin 3810310.

Measure the valve seat insert bore depth in the cylinder head.

Insert Bore Depth (Standard Insert) for Cylinder Heads Built Prior to August 21, 2006		
mm		in
9.40	MIN	0.370
9.50	MAX	0.374

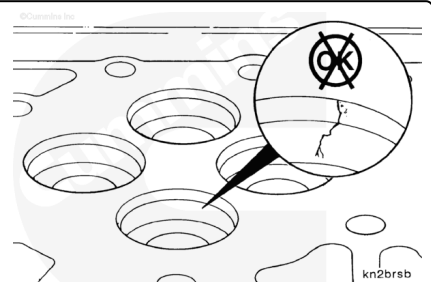


Insert Bore Depth (Standard Insert) for Cylinder Heads Built After August 21, 2006, with the Scallop		
mm		in
9.15	MIN	0.360
9.25	MAX	0.364

If the valve seat insert bore depth does **not** meet the specifications given, refer to Cylinder Head Oversize Valve Seat Insert installation in the L10 Alternative Repair Manual, Bulletin 3810310.

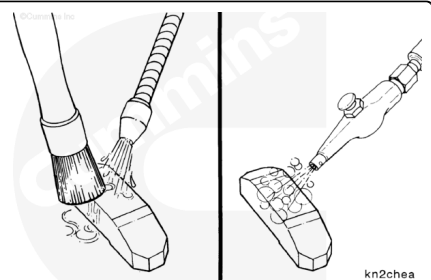
Inspect the insert bore for cracks or other damage.

If cracks or other damage is found, it is possible to repair the cylinder head by machining the insert bore for oversize valve seat inserts. Refer to the L10 Alternative Repair Manual, Bulletin 3810310.



⚠ WARNING ⚠

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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

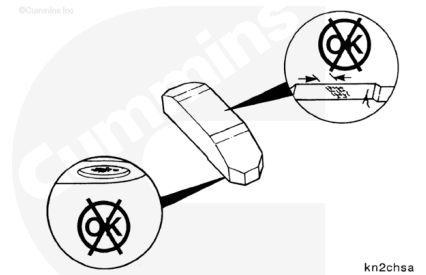
Use solvent to clean the parts.

Dry with compressed air.

Inspect the rocker lever contact pad for wear, cracks, or other damage.

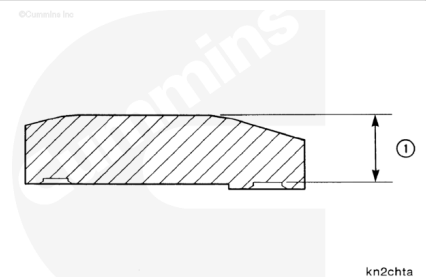
Wear in the pad contact area **must not** exceed 7 mm [0.276 in] width.

Dry with compressed air.



Measure the valve stem pocket depth from the valve stem pocket to the rocker pad face (1).

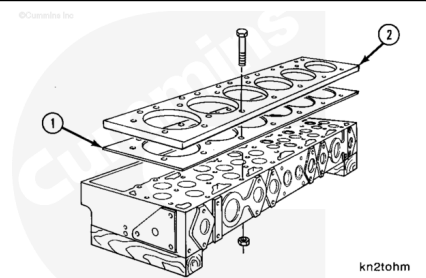
Stem Pocket to Pad Face		
mm		in
12.05	MIN	0.474
12.55	MAX	0.494



Pressure Test

⚠ CAUTION ⚠

Do not pressure-test the cylinder head with the valves and valve springs installed. Water entering the cylinder head can not be dried thoroughly and will damage the valve guides and valve stems.



Coolant Passage

Install cylinder head water test fixture, Part Number 3376082:

- Install gasket, Part Number 3376084 (1)
- Install test plate, Part Number 3376658 (2)

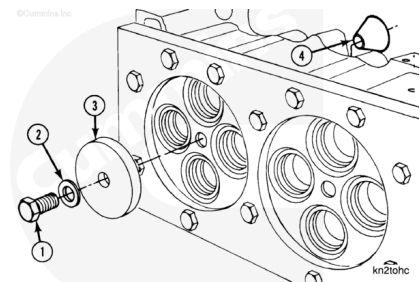
- Install the 32 cylinder head capscrews and 32 (M14 x 1.50) nuts.

Tighten the nuts.

Torque Value: 47 n•m [35 ft-lb]

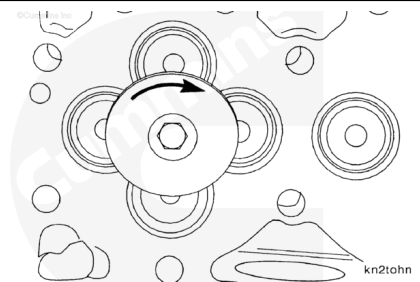
Use six injector sleeve holding tools, Part Number ST-1179.

Install the capscrew (1), flat washer (2), anvil (3), Part Number ST-179-4, and mandrel (4), Part Number ST-1179-2, in each injector bore as shown.



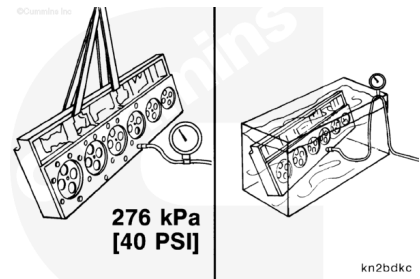
Tighten the six injector sleeve holding tool capscrews.

Torque Value: 75 n•m [55 ft-lb]



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Connect a regulated air supply hose to the test fixture plate.
Apply air pressure.

Measurements

	kpa	psi
Air Pressure	276	40

Use a nylon lifting strap and a hoist to place the cylinder head in a tank of heated water.

Completely submerge the cylinder head in the water.

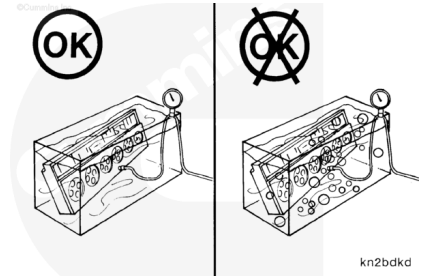
Measurements

	celsius	fahrenheit
Water Temperature	60	140

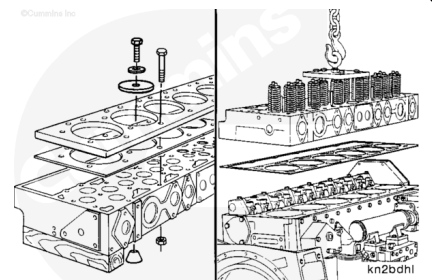
Inspect for air bubbles rising from the water.

If air bubbles are seen, replace or rebuild the cylinder head.

If a crack is found in the cylinder head, preserve the cylinder head with clean engine oil so that rust does **not** develop.



Remove the test equipment.



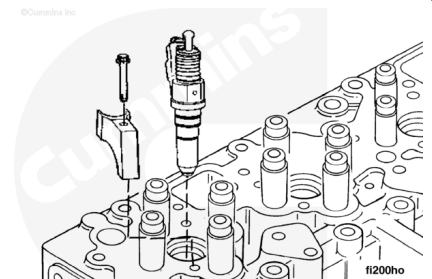
Fuel Passage

Use clean 15W-40 oil to lubricate the injector o-rings.

Install a set of injectors with new o-rings in the cylinder head.

See Section 6 for proper o-ring orientation.

Install the injector hold-down clamps and mounting capscrews.

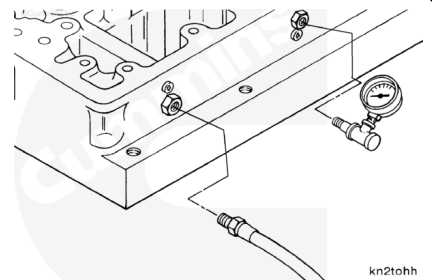


Torque Value: 75 n•m [55 ft-lb]

Install a 0-160 psi pressure gauge, Part Number 3375275, to the fuel outlet fitting.

Install an air supply line to the fuel inlet fitting.

Apply air pressure to the cylinder head.



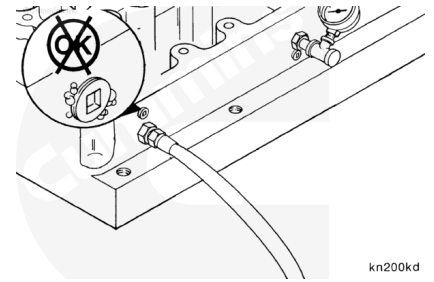
Air Pressure

kpa		psi
552	MIN	80
689	MAX	100

Close the air supply valve and inspect the fuel passages for leaks.

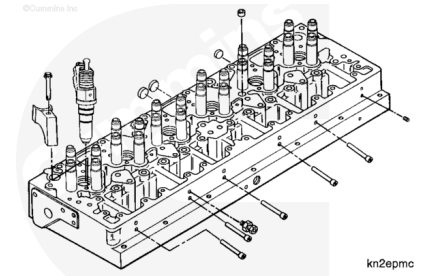
Record the air pressure on the gauge. The air pressure **must not** decrease on the gauge for at least 15 seconds.

If the air pressure decreased, use a solution of soap and water on the fuel passage areas and pipe plugs to find the leak(s).



kn200kd

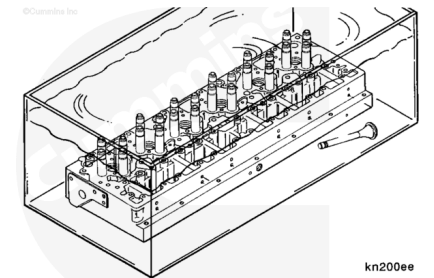
Remove the injectors from the cylinder head.



kn2epmc

Install the cylinder head and parts in a tank of cleaning solution.

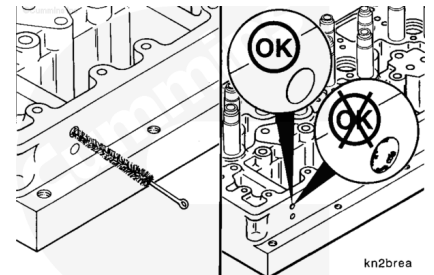
Be sure to follow the cleaning solution manufacturer's instructions when cleaning the parts.



kn200ee

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



kn2bre

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove the cylinder head and parts from the cleaning tank.

Use fuel passage cleaning brush, Part Number ST-876, to clean the fuel and oil passages with solvent. Dry with compressed air.

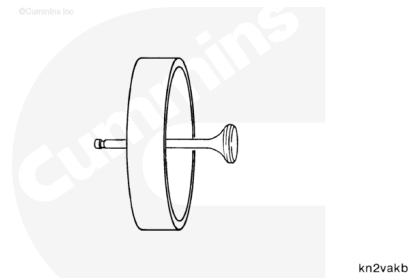
Inspect the fuel and oil passages to make sure they are clean.

Magnetic Crack Inspect

Use the magnetic particle residual method to inspect the valves for cracks.

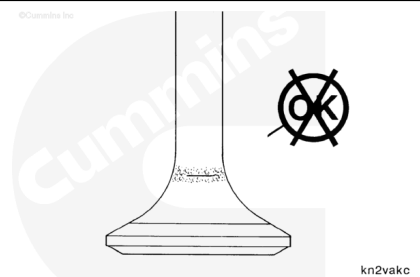
The exhaust valves contain two types of metal and **must** be inspected by the coil shot method. There will be a magnetic leakage apparent at the point where the two metals are welded together. The leakage will appear as a wide pattern of magnetic particles.

Magnetize the valves in a coil of 100 to 200 amperes.



Inspect the valves with residual Magnaglo®.

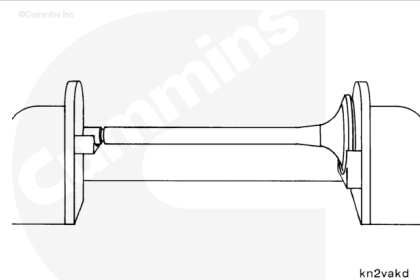
A broad fuzzy pattern will appear at the welded joint on the exhaust valves. If there is a distinct line in the pattern, the valve **must** be replaced.



The intake valves contain **only** one type of metal and **must** be magnetized and inspected in two directions. Use the Magnaglo® coil method and inspect with residual Magnaglo®, the same as for the exhaust valves.

Magnetize the intake valves again with a headshot at 500 to 700 amperes.

Inspect the valves again with residual Magnaglo®.



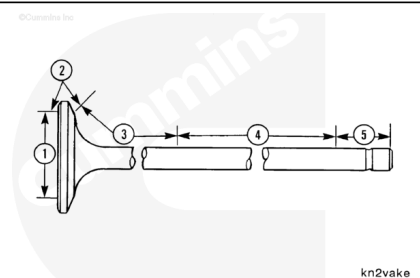
Acceptance Criteria (Exhaust and Intake Valves)

A visible crack is one which can be seen through a three power magnifying glass after the magnetic particle suspension is removed.

Area 1, no indication longer than 12.70 mm [0.500 in].

There **must** be no more than five indications or no indications closer together than 3.18 mm [0.125 in].

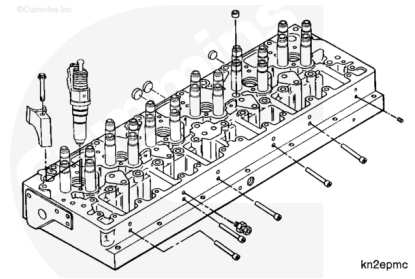
Areas 2, 3, 4, and 5 **must not** have any magnetic indications or visible indications.



Remove all magnetism and clean the acceptable valves.

Remove the pipe plugs, fuel passage plugs, and fuel fittings from the cylinder head.

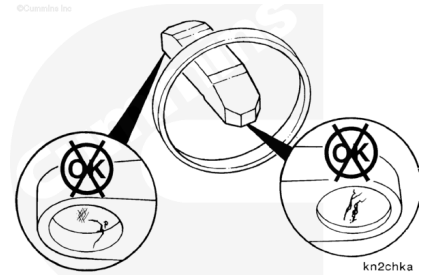
Use a dent puller to remove the expansion plugs.



Magnetize the crosshead and inspect it with residual Magnaglo®.

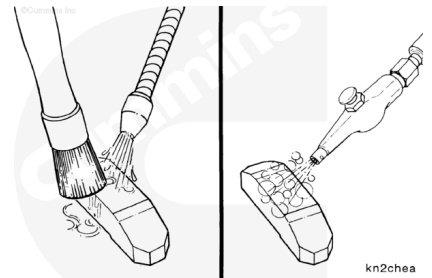
Cracks will be indicated by a bright line. The primary location for cracks is the valve stem pocket area.

If cracks are found, the crosshead **must** be replaced.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendation for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

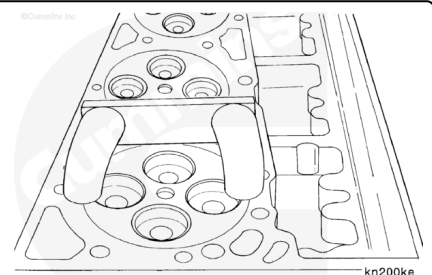
Remove all magnetism.

Use solvent to clean the crossheads.

Dry with compressed air.

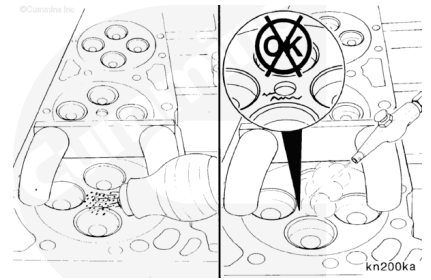
Use magnetic crack detector, Part Number ST-1166, to find cracks in the combustion face, in areas around the valve ports, and in the injector bores.

Place the magnetizing head on the combustion face as shown to check for cracks that run across the cylinder head.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Fill the powder spray bulb one-third full with metal powder.

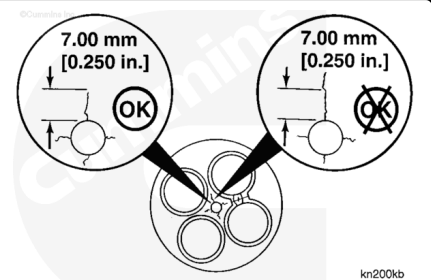
Spray the powder lightly onto the combustion face.

Use compressed air to remove the excess metal powder. Air Pressure: 276 kPa [40 psi]

The powder will remain in the cracks, if present, and show as a white line.

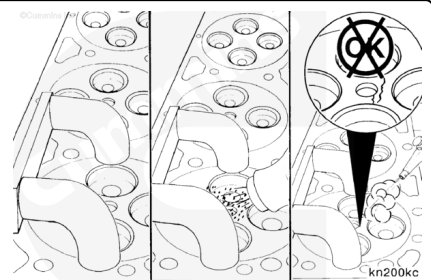
Inspect the cylinder head for cracks in the combustion face.

If any cracks around the injector bore are longer than 7 mm [0.250 in.], the cylinder head **must** be replaced.



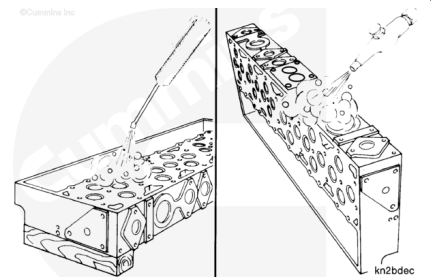
Place the magnetizing head on the combustion face as shown to check for cracks that run lengthwise on the cylinder head.

Repeat the procedure as outlined above.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendation for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove all magnetism and use solvent to clean the cylinder head. Dry with compressed air.

The cylinder head **must** be thoroughly cleaned after using the magnetic crack detector to remove all of the iron fragments.

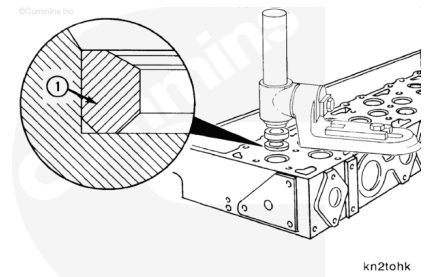
Assemble

Install the base and swivel of the valve seat insert tool, Part Number ST-275, on the cylinder head to guide the valve seat driver. Install valve guide arbor, Part Number ST-804-1.

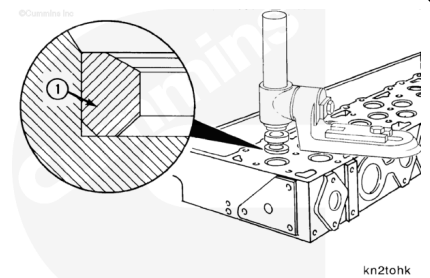
The insert chamfer (1) **must** be installed toward the bottom of the counterbore.

Use valve seat driver, Part Number 3376105, to drive the valve seat insert into the counterbore.

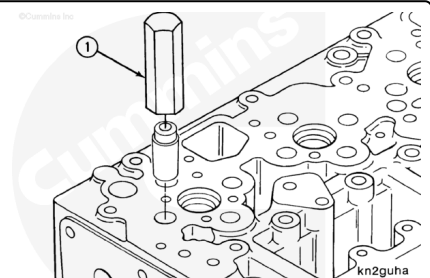
Make sure the insert is at the bottom of the counterbore.



Use valve seat grinding machine, Part Number ST-685, and valve guide arbor set, Part Number 804, to grind the new valve seat inserts.

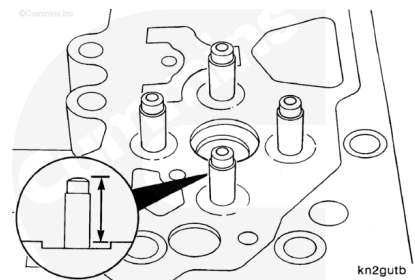


Use valve guide driver (1), Part Number 3823947, to install the new valve guides.



Measure the valve guide installed height above the cylinder head top deck surface.

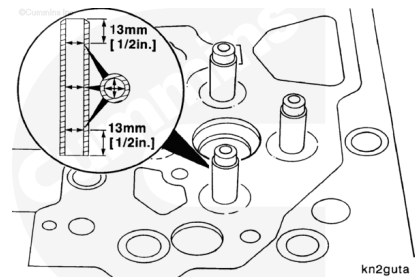
See the Clean and Inspect for Reuse section of this procedure for specifications.



Measure the valve guide inside diameter.

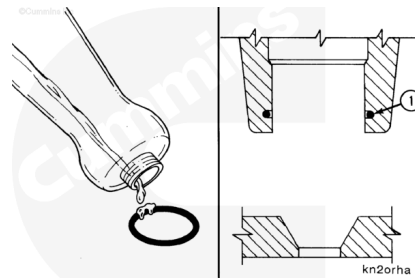
See the Clean and Inspect for Reuse section of this procedure for specifications.

If the valve guide is **not** within the specifications given, the valve guide **must** be replaced.



Use vegetable oil to lubricate the injector sleeve o-rings.

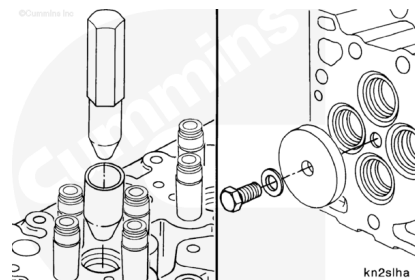
Install the o-ring (1) in the groove of the injector sleeve.



Use injector sleeve driver, Part Number ST-1227, to install the injector sleeve into the cylinder head. Remove the sleeve driver.

Install injector sleeve holding tool, Part Number ST-1179, in the injector sleeve.

Torque Value: 50 n•m [37 ft-lb]

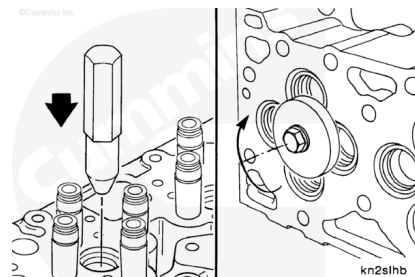


Install the injector sleeve driver.

Hit the driver with a mallet to seat the sleeve in the bore.

Remove the driver and tighten the injector sleeve holding tool capscrew again.

Torque Value: 50 n•m [37 ft-lb]

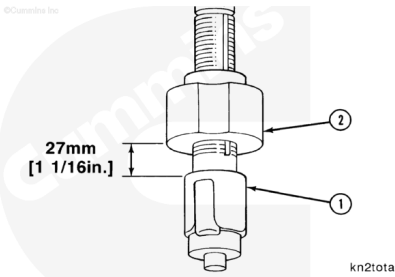


Use injector sleeve expander, Part Number ST-880, to expand the upper section of the injector sleeve.

Adjust the expander roller edge (1) and collar (2) to the clearance specified.

Measurements

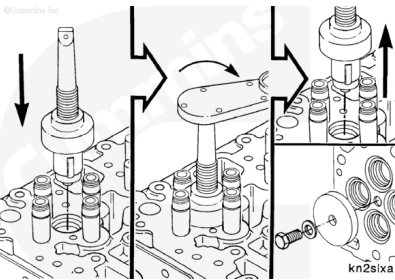
	mm	in
Clearance	27	1.06



Install the expander into the injector sleeve and turn the mandrel with a 1/2-in socket and an inch-pound torque wrench.

Torque Value: 9 n•m [80 in-lb]

Remove the expander and holding tool from the injector sleeve.



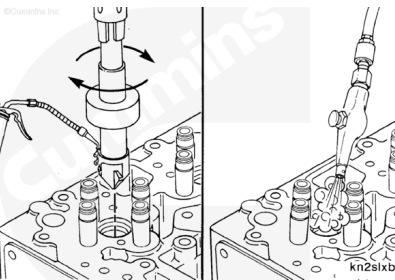
Use pilot, Part Number ST-379-2, holder, Part Number ST-884-1, and injector seat cutter, ST-844-3, to cut the injector seat.

Install the injector sleeve cutter and the cylinder head in a drill press.

Use a cutting oil to lubricate the cutter head.

Carefully machine the injector sleeve until the sealing area is smooth.

Clean the metal particles from the injector sleeve bore and fuel passages.

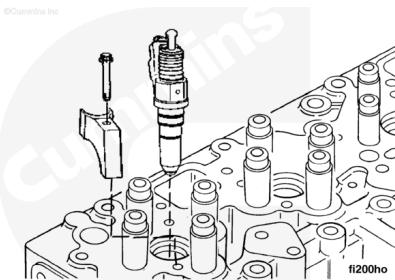


⚠ CAUTION ⚠

Support the cylinder head in the Part Number ST-583 head holding fixture to reduce the possibility of damage to the injector tip that protrudes from the combustion face.

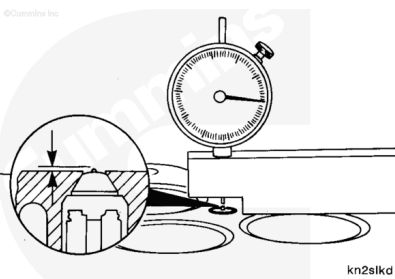
Install the injectors into the cylinder head without the o-rings.

Torque Value: 75 n•m [55 ft-lb]



Turn the cylinder head over and use depth gauge, Part Number 3823495, to measure the injector tip protrusion.

Injector Tip Protrusion (New Injector Sleeve)		
mm		in
2.28	MIN	0.090



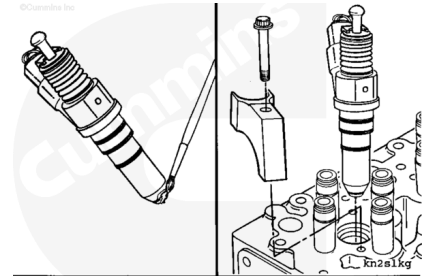
Injector Tip Protrusion (New Injector Sleeve)

mm		in
2.54	MAX	0.100

Remove the injectors and apply a bluing compound to the outside diameter of the injector at the injector seat area.

Install the injectors into the cylinder head without the o-rings.

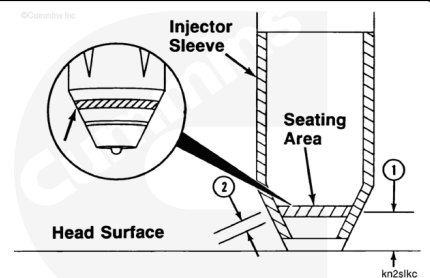
Torque Value: 75 n•m [55 ft-lb]



The bluing pattern of the injector seating area (1) **must** be approximately 13 mm [0.50 in] from the cylinder head surface.

The injector bore seating width (2) **must** be a minimum of 1.52 mm [0.060 in].

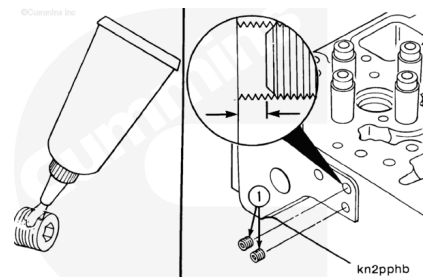
Inspect the cylinder head for leaks after the new injector sleeves have been installed. See the Pressure Test section of this procedure.



Apply a coat of cup plug sealant, Part Number 3375068, to the pipe plug threads. Install the pipe plugs.

The pipe plugs (1) located in the front of the cylinder head **must** be installed below the surface of the cylinder head.

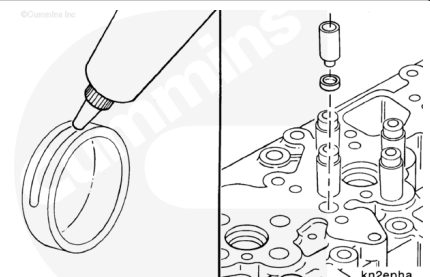
Torque Value: 10 n•m [89 in-lb]



Apply a coat of cup plug sealant, Part Number 3375068, to the outside diameter of the expansion plugs.

Use expansion plug drivers, Part Numbers 3822372, 3376058, and 3376816, to install the expansion plugs in the cylinder head.

The correct expansion plug driver will install the expansion plugs to the correct depth in the cylinder head.



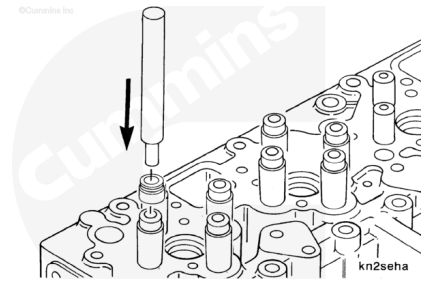
⚠ CAUTION ⚠

Do not use excessive force or the seal will be damaged.

Use valve seal driver, Part Number 3823946, to install the valve seals on the valve guides.

Position the seal on the tool arbor.

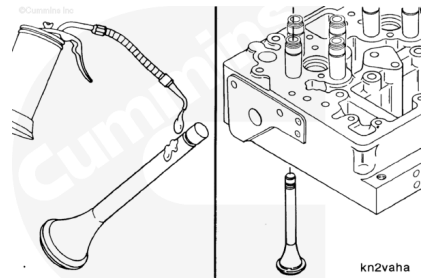
Insert the arbor into the top of the guide and tap the tool with a plastic mallet to seat the bottom of the seal against the machined step on the valve guide.



Use clean 80W, or higher, gear lube with EP additives to lubricate the valve stems.

Install the valves in the valve guides.

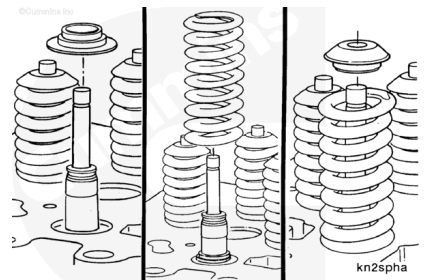
After the valves are installed, place the cylinder head on a flat surface that will **not** damage the cylinder head surface.



Install the valve spring wear plates.

Install the valve springs.

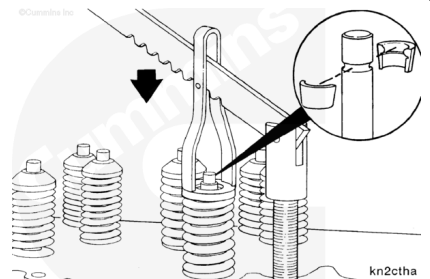
Install the valve spring retainers.



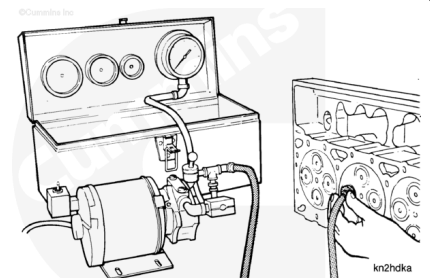
Use valve spring compressor, Part Number ST-448, compressor thread adapter, Part Number 3376850, and compressor adapter, Part Number 3376851, to compress the valve springs.

Always use new collets when rebuilding the cylinder head.

Install the new valve spring retainer collets.



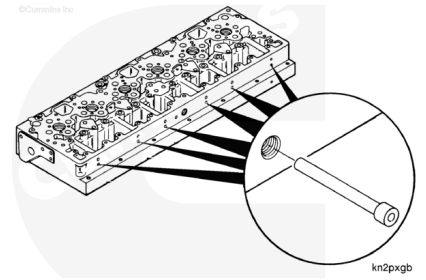
Use vacuum tester, Part Number ST-1257, with vacuum cup, Part Number 3376100, to vacuum test the valve seating. See Cylinder Head - Vacuum Test Valve Seating.



Install

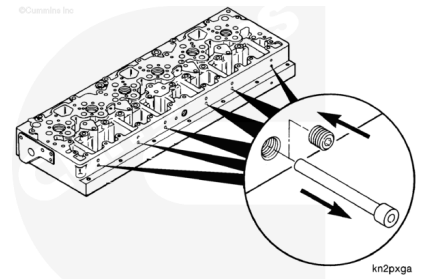
A plastic insert is installed in all fuel inlet passages on new and ReCon® cylinder heads.

The fuel inlet passages are the lower row of drillings.



Depending on the fuel plumbing option, one fuel inlet passage insert **must** be removed to allow fuel to flow to the injectors. If the insert is **not** removed, the engine will **not** start.

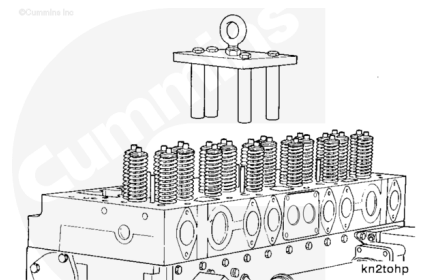
Remove the pipe plug and use a sharp object to pull the insert from the hole.



Install the cylinder head lifting bracket, Part Number 3822479, with two rocker lever support mounting capscrews and two long rocker housing mounting capscrews.

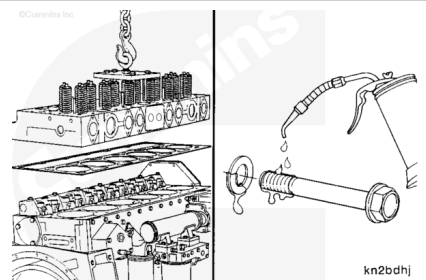
Tighten the capscrews.

Torque Value: 47 n•m [35 ft-lb]



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not drop the cylinder head on the cylinder head gasket. The gasket material can be damaged.

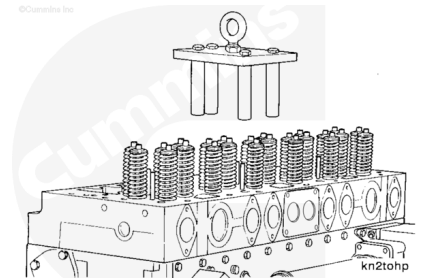
Install a new cylinder head gasket.

Use a hoist or hydraulic arm and install the cylinder head.

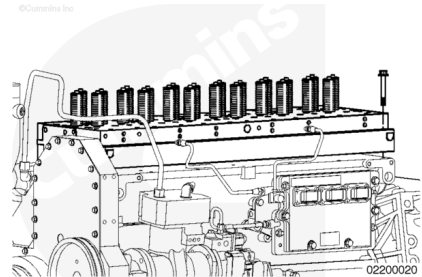
Use clean 15W-40 oil to lightly coat the cylinder head capscrew threads and bottom of the flange.

Allow the excess oil to drain from the capscrew threads.

Remove cylinder head lifting bracket, Part Number 3822479.

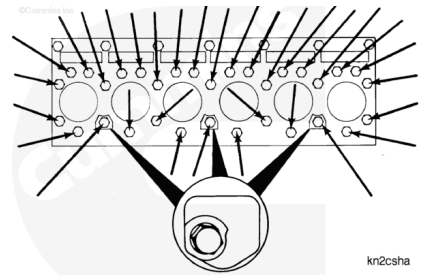


Install the seven hex flange head capscrews with new sealing washers on the fuel pump side of the engine.



Install the remaining 32 cylinder head capscrews in the sequence shown.

Be sure to install the three short capscrews inside the intake ports.

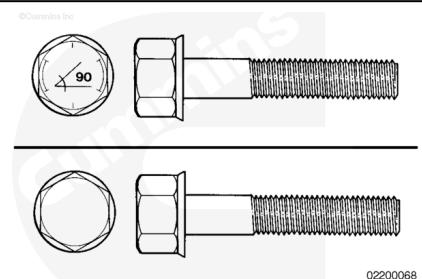


All 32 capscrews, old and new, **must** be rotated 180 degrees after they have been tightened according to torque step 2. Make a mark on the cylinder head as a reference. Rotate the capscrew more than two flats and but less than three flats.

After tightening according to torque step 2, mark the cylinder head as a reference.

When using torque plus angle, the tolerance on the 180-degree angle of rotation is between two and three flats. If the capscrew is unintentionally rotated beyond three flats, do **not** loosen the capscrew. The clamp load is still acceptable. However, rotating the capscrew beyond three flats reduces the number of potential reuses.

Note : This torque procedure is applicable to both old and new capscrews.

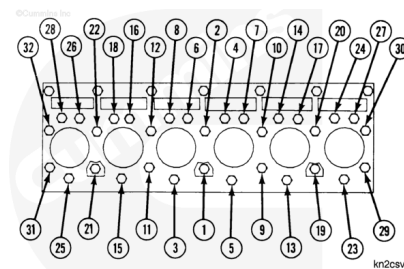


Tighten all the capscrews in the sequence shown to the specified values:

Torque Value:

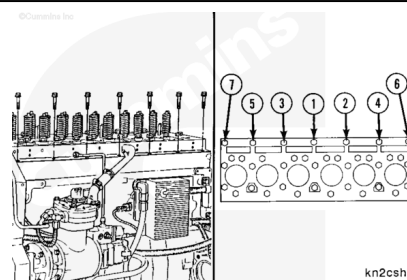
1. 68 n•m [50 ft-lb]
2. 136 n•m [100 ft-lb]
3. Rotate 180 degrees.

Note : Rotate at least two flats, but not more than three flats.



Tighten the seven capscrews in the fuel pump side of the cylinder head in the sequence shown.

Torque Value: 68 n•m [50 ft-lb]



Finishing Steps

With EGR

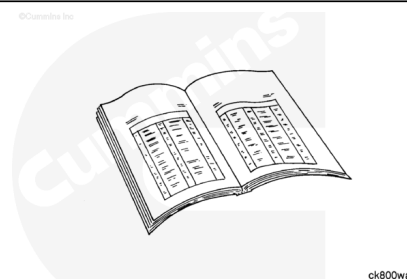
Note : If the cylinder head, cylinder head gasket, or the cylinder head injector sleeve is being replaced due to a malfunction that caused an internal coolant leak, the external crankcase breather element must be replaced if the unit is equipped with an external crankcase breather element. Refer to Procedure 003-019 in Section 3.

(/qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html)

Note : If a malfunction resulted in coolant entering the exhaust system, the aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)

Note : If a malfunction resulted in oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Reference the Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600.

Note : Use the following procedure if the piston(s) is(are) being replaced due to damaged exhaust valves,



intake valves, pistons, or other debris generating failures. Refer to the Prevention of Turbocharger Damage After Engine Mechanical Issue, Service Bulletin 4326040
(/qs3/pubsys2/xml/en/bulletin/4326040.html)

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

- Install the refrigerant compressor and mounting bracket.
- Install the CM870 ECM mounting bracket. Refer to Procedure 019-031
(/qs3/pubsys2/xml/en/procedures/88/88-019-031.html)
in Section 19 in the CM870 Electronic Control System, ISM Engines Troubleshooting and Repair Manual, Bulletin 4021381.
- Connect the ECM connector to the ECM. Refer to Procedure 006-006 in Section 6.
(/qs3/pubsys2/xml/en/procedures/35/35-006-006-tr.html)
- Install the fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-036-tr.html)
- Install the cooling fan. Refer to Procedure 008-040 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-040-tr.html)
- Install the fuel lines to the cylinder head. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/35/35-006-024-tr.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-007-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-

010-033-tr.html)

- Install the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-013-tr.html)
- Install the cylinder head vent lines.
- Install the EGR cooler. Refer to Procedure 011-019 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-019-tr.html)
- Install the EGR valve. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)
- Install the EGR cooler coolant return and supply tubes. Refer to Procedure 011-019 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-019-tr.html)
- Install the EGR cooler vent lines. Refer to Procedure 011-031 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Install the water manifold. Refer to Procedure 008-061 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)
- Install the EGR connection tube. Refer to Procedure 011-025 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)
- Install the EGR mixer and air horn assembly. Refer to Procedure 011-021 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html)
- Install the EGR differential pressure sensor and exhaust pressure sensor tubes. Refer to Procedure 011-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html) Refer to Procedure 011-027 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)
- Connect the exhaust pressure sensor to the harness. Refer to Procedure 019-376 in Section 19 in the CM870 Electronic Control System, ISM Engines Troubleshooting and Repair Manual, Bulletin 4021381.
(/qs3/pubsys2/xml/en/procedures/88/88-019-376.html)
- Connect the EGR differential pressure sensor harness and sensor. Refer to Procedure 019-370 in Section 19 in the CM870 Electronic Control System, ISM Engines

Troubleshooting and Repair Manual, Bulletin 4021381.
(/qs3/pubsys2/xml/en/procedures/88/88-019-370.html)

- Connect the air compressor air supply tube to the EGR mixer. Refer to Procedure 012-109 in Section 12.
(/qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html)
- Install the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/35/35-006-026-tr.html)
- Install the crossheads. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html)
- Install the rocker lever assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html)
- Install the push tubes or push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)
- Install the internal wiring harness to the engine.
- Connect the Deutsch™ 2-pin connectors to the internal engine wiring harness to the fuel injectors.
- Install the internal engine wiring harness holding clamp capscrews.
- Install the locking capscrew to the Metri-Pack™ 12-pin connector at the front of rocker lever housing, and connect the connector. Connect the engine brake oil feed line, if equipped.
- Install the engine brakes, if applicable. Refer to Procedure 020-024 in Section 20.
(/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Install the crankcase breather, if applicable. Refer to Procedure 003-001 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-001-tr.html)
- Connect the air piping to the intake manifold and turbocharger. Refer to Procedure 010-023 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-023-tr.html) Refer to Procedure 010-033 in Section 10.

(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine to normal operating temperature and check for leaks.

Without EGR

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

- Install the fan hub and fan. Refer to Procedure 008-036 in Section. (/qs3/pubsys2/xml/en/procedures/35/35-008-036-tr.html) Refer to Procedure 008-040 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-040-tr.html)
- Install the brackets and clamps that were removed.
- Install the fuel lines and air compressor tube to the cylinder head.
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-007-tr.html)
- Install the rocker lever housing. Refer to Procedure 003-013 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-013-tr.html)
- Install the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/35/35-006-026-

tr.html)

- Install the rocker lever assemblies. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html)
- Install the internal engine wiring harness on the engine.
- Connect the 12-pin connector of the internal engine harness to the 12-pin connector of the external engine harness.
- Install the locking capscrew (Torx™ size T-25). Tighten the capscrew.
- Install the holding clamps for the internal engine harness.
- Connect the Deutsch™ 2-pin connectors to the internal engine harness to the fuel injectors.
- Install the crossheads. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-009-tr.html)
- Install the push tubes and push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)
- Install the engine brakes, if equipped. Refer to Procedure 020-024 in Section 20. (/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Connect the air piping to the turbocharger and the intake manifold. See the manufacturer's specifications for the correct torque value.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Connect the battery cable. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine to normal operating temperature and check for leaks.

